Cardiogenic shock

Definition and staging

- SBP < 90 (or vasopressor) + hypoperfusion (lactate > 2, oliguria, cold) from cardiac failure; CI < 2.2
- **SCAI stages A–E**: at risk, beginning, classic, deteriorating, extremis

Immediate management

- Emergency revascularisation of culprit only (CULPRIT-SHOCK)
- Noradrenaline for MAP 65; dobutamine (or milrinone) if low CO; avoid excessive fluid
- Echo, PA catheter in refractory shock; ventilation, RRT if needed

Mechanical support

- IABP-SHOCK II: no mortality benefit; **DanGer Shock 2024: Impella CP ↓ mortality in STEMI shock**
- ECLS-SHOCK: VA-ECMO no benefit, more bleeding; shock team decision

Key point: Revascularise the culprit fast; use devices selectively with a shock team.

Mechanical circulatory support: IABP, Impella, VA-ECMO

Devices

- **IABP**: inflates in diastole (↑ coronary flow), deflates before systole (↓ afterload); +0.5 L/min
- **Impella**: axial pump LV → aorta, unloads LV, 2.5–5.5 L/min
- **VA-ECMO**: full biventricular + respiratory support; ↑ LV afterload

Choice

- LV failure with adequate oxygenation: Impella; biventricular failure/arrest/hypoxia: VA-ECMO
- Combine ECMO + Impella (ECPELLA) to vent the LV

VA-ECMO complications

- Bleeding, limb ischaemia (distal perfusion cannula), stroke, haemolysis, infection
- LV distension/pulmonary oedema; **Harlequin (north–south) syndrome**: right radial ABG

Key point: Match the device to the failing ventricle and oxygenation.

Notes: cardiology p.7

Acute heart failure

Classification

- Warm/cold (perfusion) × wet/dry (congestion): warm-wet most common
- Phenotypes: acute decompensated, pulmonary oedema, RV failure, cardiogenic shock

Management

- Warm-wet: IV furosemide (≥ 1–2.5× home dose), nitrates if SBP > 110, **CPAP/NIV**
- Cold-wet: inotrope ± vasopressor, then diuretics; find precipitant (CHAMPIT)
- Start/continue GDMT once stable (STRONG-HF)

Inotropes and vasodilators

- Dobutamine/milrinone: short-term only, arrhythmia and mortality signal
- Levosimendan (calcium sensitiser): no mortality benefit (SURVIVE, LIDO)

Key point: Treat congestion with diuretics + NIV; inotropes only for low output.

Notes: cardiology p.10

Myocardial infarction and NSTEMI

Classification (4th universal definition)

- Type 1 plaque rupture; type 2 supply–demand mismatch (sepsis, anaemia, tachycardia)
- Type 3 sudden death; 4 PCI-related; 5 CABG-related; myocardial injury = troponin without ischaemia

NSTEMI management

- DAPT, anticoagulation, statin, beta-blocker; GRACE score
- Invasive within 24 h if high risk; immediate (< 2 h) if unstable/refractory pain

Thrombocytopenia

- Heparin (HIT), GP IIb/IIIa inhibitors (acute, within hours), sepsis, dilution
- Use 4T score; switch to bivalirudin/argatroban if HIT

Complications of STEMI

- Arrhythmia, heart failure/shock, mechanical rupture, pericarditis, LV thrombus

Key point: In ICU most troponin rises are type 2 or injury: treat the cause.

Notes: cardiology p.17

Mechanical complications of MI

Types (days 3–7)

- Ventricular septal rupture: new harsh murmur, step-up in O2 saturation (RV)
- Papillary muscle rupture: acute MR, flash pulmonary oedema, soft/absent murmur
- Free wall rupture: tamponade, PEA, sudden death

Haemodynamic management

- Afterload reduction (nitroprusside if BP allows), IABP/Impella, VA-ECMO bridge
- Inotropes, ventilation, avoid excessive vasopressors

Surgery and prognosis

- Urgent surgery for all; delay (bridge on MCS) may improve tissue repair
- Mortality 40–80%; percutaneous VSR closure option

Key point: New murmur + shock after MI = mechanical complication: echo now.

Post-cardiac surgery tamponade

Diagnosis

- Hypotension, tachycardia, rising CVP, falling then stopped drain output, low UO
- Echo may show localised clot (TOE); equalised pressures; pulsus paradoxus

Immediate management

- Fluid, inotrope, avoid high PEEP; correct coagulopathy
- **Re-sternotomy** (bedside if arrest); pericardiocentesis for effusion in other settings

Tamponade vs constriction

- Tamponade: pulsus paradoxus, Y descent absent, diastolic collapse
- Constriction: Kussmaul's sign, prominent Y descent, pericardial knock, thick pericardium

Key point: Sudden fall in drain output with shock after cardiac surgery = tamponade.

Electrical storm

Definition

- ≥ 3 episodes of sustained VT/VF or appropriate ICD shocks in 24 h
- Precipitants: ischaemia, heart failure, electrolytes, drugs (QT), thyroid, infection

Acute management

- Defibrillation/ACLS, correct K > 4.5, Mg > 1, treat ischaemia
- **Amiodarone + beta-blocker (propranolol)**; lidocaine; sedation/intubation
- Stellate ganglion block, catheter ablation, MCS as bridge

Polymorphic VT

- Long QT (torsades): magnesium 2 g, stop drugs, overdrive pacing/isoprenaline
- Normal QT: treat ischaemia (angiography)

Key point: Sedation and sympathetic blockade calm the electrical storm.

New-onset AF in critical illness

Causes and significance

- Sepsis, catecholamines, electrolytes, hypovolaemia, inflammation, atrial stretch, surgery
- Associated with ↑ mortality, stroke, longer stay

Rate/rhythm control

- Correct K/Mg, reduce catecholamines where possible
- **Unstable: synchronised DC cardioversion**; amiodarone first-line in shock
- Beta-blocker (esmolol/landiolol) if BP allows; digoxin adjunct; avoid diltiazem in low EF

Anticoagulation

- CHA2DS2-VASc vs bleeding risk; often reverts in ICU; consider if persists > 48 h
- Weigh bleeding (HAS-BLED) and procedures

Key point: Fix the trigger first: many ICU AF episodes revert spontaneously.

Hypertensive emergency

Definitions

- Emergency: severe BP (often > 180/120) + acute organ damage; urgency: no organ damage
- Organs: brain, heart, aorta, kidney, eyes, placenta

BP targets

- **Reduce MAP ≤ 25% in first hour**, then to 160/100 over 2–6 h, normal over 24–48 h
- Exceptions: aortic dissection (SBP < 120 within 20 min), ICH (SBP 140), stroke for lysis (< 185/110), eclampsia

IV agents

- Labetalol, nicardipine, clevidipine, esmolol, GTN (coronary/pulmonary oedema)
- Sodium nitroprusside: potent, cyanide toxicity; phentolamine for catecholamine crisis

Key point: Lower BP gradually except in dissection: too fast causes ischaemia.

Posterior reversible encephalopathy syndrome (PRES)

Pathophysiology

- Failed cerebral autoregulation → vasogenic oedema; endothelial dysfunction
- Posterior circulation (poor sympathetic innervation) most affected

Causes and triggers

- Hypertension, eclampsia, renal failure, **calcineurin inhibitors (tacrolimus)**, chemotherapy, SLE, sepsis

Management

- Lower BP gradually; stop offending drug; treat seizures (levetiracetam)
- MRI: parieto-occipital vasogenic oedema (FLAIR); magnesium in eclampsia

Prognosis

- Usually reversible in days–weeks; poor with haemorrhage, cytotoxic oedema, delayed treatment

Key point: Headache + seizures + visual loss with high BP or tacrolimus → MRI for PRES.

Acute aortic syndrome

Classification

- Stanford A (ascending, surgery) / B (descending, mostly medical); DeBakey I–III
- Includes intramural haematoma and penetrating aortic ulcer

Medical management

- **IV beta-blocker first (esmolol/labetalol) → HR < 60, then SBP < 120**
- Analgesia; vasodilator after beta-blockade

Definitive

- Type A: emergency surgery; type B complicated (malperfusion, rupture, refractory pain/BP): TEVAR

Complications

- Tamponade, AR, MI, stroke, mesenteric/renal/limb ischaemia, paraplegia

Key point: Rate control before vasodilators in aortic dissection.

Phaeochromocytoma crisis

Features and diagnosis

- Paroxysmal headache, sweating, palpitations, hypertension; crisis with surgery/anaesthesia
- **Plasma free or urinary metanephrines**, then CT/MRI, MIBG/PET

Crisis management

- **Alpha-blockade first** (phentolamine bolus, then infusion); magnesium, nicardipine
- Beta-blocker only after alpha-blockade (avoid unopposed alpha)
- After removal/stopping trigger: profound hypotension → fluids, vasopressors; hypoglycaemia

Elective preparation

- Phenoxybenzamine or doxazosin 10–14 days, then beta-blocker; liberal salt and fluid

Key point: Never beta-block before alpha-block in phaeochromocytoma.

ECPR

Selection

- Witnessed arrest, bystander CPR, shockable rhythm, age < 70, no-flow < 5 min, low-flow < 60 min
- Reversible cause, no major comorbidity, EtCO2 > 10, arrival within 60 min

Evidence

- **ARREST 2020: survival 43% vs 7%** (stopped early)
- Prague OHCA: no significant benefit; INCEPTION: no benefit (system dependent)

Contraindications

- Unwitnessed, prolonged no-flow, terminal illness, aortic dissection, severe AR, frailty

Post-ECPR care

- Coronary angiography, LV venting, limb perfusion, temperature control, neuroprognostication

Key point: ECPR works in selected patients within well-trained systems.

High-quality CPR, reversible causes and POCUS

High-quality CPR

- 100–120/min, 5–6 cm, full recoil, pauses < 10 s, rotate every 2 min
- Adrenaline 1 mg every 3–5 min; amiodarone 300 mg after 3rd shock
- EtCO2 > 20 mmHg good CPR; sudden rise suggests ROSC

Reversible causes (4 Hs, 4 Ts)

- Hypoxia, hypovolaemia, hypo/hyperkalaemia, hypothermia
- Thrombosis (coronary/PE), tension pneumothorax, tamponade, toxins

POCUS during arrest

- Only in pulse checks (< 10 s); identifies tamponade, RV dilatation, hypovolaemia, pseudo-PEA

Key point: Minimise pauses: every second without compressions lowers survival.

Notes: cardiac arrest p.1

Post-cardiac arrest syndrome and angiography

Emergency angiography

- STEMI after ROSC: immediate; no STEMI: delayed approach as good (COACT, TOMAHAWK)

Post-cardiac arrest syndrome

- Brain injury, myocardial dysfunction, systemic ischaemia–reperfusion (SIRS-like), persistent cause

Targets

- MAP ≥ 65 (no benefit from higher: BOX), SpO2 94–98%, normocapnia (PaCO2 4.5–6)
- Fever avoidance ≤ 37.7 for 72 h, glucose 6–10, treat seizures

Key point: Post-arrest care: avoid hypoxia, hypotension, hyperthermia.

Targeted temperature management

Evolution of evidence

- 2002 HACA & Bernard: 32–34 °C improved outcome
- TTM-1 2013: 33 = 36 °C; HYPERION 2019: 33 benefit in non-shockable
- **TTM-2 2021: 33 °C = normothermia with fever control**
- Current: prevent fever (≤ 37.7 °C) for at least 72 h

Practical delivery

- Surface or intravascular devices with feedback; sedation; control shivering (BSAS)
- Monitor K, glucose, coagulation, bradycardia

Neuroprotection

- ↓ Metabolic rate, excitotoxicity, free radicals, apoptosis, BBB disruption

Key point: Fever is harmful after cardiac arrest: control temperature, not depth.

Neuroprognostication after cardiac arrest

Timing

- **≥ 72 h after ROSC** after sedation is cleared; multimodal

Predictors of poor outcome (ERC 2021)

- Absent pupillary + corneal reflexes ≥ 72 h; bilateral absent N20 SSEPs
- NSE > 60 µg/L at 48–72 h; status myoclonus ≤ 72 h; malignant EEG
- CT diffuse oedema (low GWR); MRI diffusion restriction

Family communication

- Honest, consistent, uncertainty explained, regular meetings, involve team

Key point: Never prognosticate from a single test: use ≥ 2 concordant findings.

Mechanical CPR, survival factors and arrest drugs

Mechanical devices

- LUCAS, AutoPulse: no better survival (LINC, PARAMEDIC, CIRC)
- Useful in transport, PCI lab, prolonged CPR, ECPR bridge

Survival factors

- Witnessed arrest, bystander CPR, early defibrillation (public AED), shockable rhythm
- Short time to ROSC, good post-arrest care in cardiac arrest centres

Drugs

- Adrenaline: ↑ ROSC and survival, not good neurology (PARAMEDIC2)
- Amiodarone/lidocaine ↑ survival to admission (ALPS); no routine calcium, bicarbonate

Key point: Chain of survival: early CPR and early shock matter most.

Notes: cardiac arrest p.3

Bradyarrhythmia and AV block

Causes

- Ischaemia (inferior MI), drugs (beta-blockers, CCB, digoxin), hyperkalaemia, hypothyroidism
- Hypoxia, raised ICP, hypothermia, vagal stimulation

Immediate management

- **Atropine 500 µg IV** (max 3 mg); adrenaline/isoprenaline infusion
- Transcutaneous then transvenous pacing; treat cause

AV block types

- 1st degree: PR > 200 ms; Mobitz I: progressive PR prolongation
- Mobitz II: sudden dropped QRS (needs pacing); 3rd: AV dissociation

Key point: Mobitz II and complete heart block need pacing.

Notes: CVS 3 p.10

Fluid responsiveness and haemodynamic monitoring

Why CVP fails

- Static pressure ≠ volume; affected by RV function, PEEP, valves, IAP

Dynamic tests

- **PLR: ↑ CO ≥ 10%** (works in AF/spontaneous breathing)
- PPV/SVV > 12–13% (need controlled ventilation, Vt ≥ 8, sinus rhythm)
- End-expiratory occlusion ↑ CO 5%, mini-fluid challenge, IVC variation

Monitoring tools

- Echo: bedside, function; PiCCO: CO, EVLW, GEDV; PA catheter: RV failure, PAP, SvO2
- Pulse contour devices for continuous CO

Key point: Fluid only if responsive and still hypoperfused.